

NUTRIFLEX TECHNICAL REPORT

1. EXECUTIVE SUMMARY

The **NUTRIFLEX** project is a comprehensive AI-powered application designed to provide personalized nutrition and health management solutions. The goal of the project is to assist individuals in optimizing their diet based on their unique biological and emotional needs, including gut health, mood, and overall well-being. By combining artificial intelligence (AI) and user input, NUTRIFLEX offers tailored meal plans, calorie predictions, mood-enhancing food recommendations, and a gamified approach to motivate users to maintain a healthy lifestyle.

The problem addressed by NUTRIFLEX is the growing challenge people face in maintaining a balanced and healthy diet. Modern lifestyles, characterized by fast-paced routines and unhealthy food choices, contribute to poor physical and mental health. Many individuals struggle to find the right balance between their nutritional intake and their health goals, whether it's weight loss, muscle gain, or general wellness. Furthermore, gut health, which plays a critical role in overall health, is often neglected in conventional meal planning applications. Mood fluctuations also significantly impact people's food choices, leading to stress eating or energy dips.

NUTRIFLEX solves these problems by offering a holistic, AI-driven solution that adapts to users' unique needs. By using machine learning models, NUTRIFLEX predicts the calorie content of meals and creates personalized meal plans based on user-specific data such as Basal Metabolic Rate (BMR) and Total Energy Expenditure (TEE). The application also provides mood-based meal suggestions, helping users manage their emotional state through food choices that boost energy, focus, and relaxation.

One of the most innovative features of NUTRIFLEX is its **gut health** functionality. By allowing users to input microbiome data, the app generates personalized meal recommendations designed to improve gut health, which in turn enhances overall physical and mental well-being. This focus on gut health distinguishes NUTRIFLEX from traditional meal planners.

In addition, NUTRIFLEX integrates gamification to keep users engaged and motivated. Users can set health goals, track their progress, and earn rewards, making the health journey enjoyable and sustainable.

2. INTRODUCTION

Personalized nutrition management is becoming increasingly vital as individuals recognize that a one-size-fits-all approach to health and diet no longer works. Each person has unique nutritional requirements based on their metabolic rate, lifestyle, activity levels, and even genetic factors. However, many people still struggle to achieve and maintain balanced diets tailored to their specific needs. This struggle is compounded by limited knowledge about how certain foods can affect gut health and mood. Consequently, individuals often make poor dietary choices that lead to weight gain, fatigue, mood swings, and other health-related issues.

Gut health is a crucial, yet frequently overlooked, component of overall well-being. The human gut is home to trillions of microorganisms that play a pivotal role in digestion, nutrient absorption, immune function, and even mental health. An imbalance in gut microbiota can lead to various health problems such as digestive disorders, weakened immunity, and chronic inflammation. Furthermore, recent studies have established a strong connection between gut health and mood regulation, with the gut often referred to as the "second brain." Certain bacteria in the gut influence neurotransmitter levels, which can impact feelings of happiness, anxiety, and depression. Thus, optimizing gut health can significantly enhance physical and mental well-being.

Mood-based recommendations are also vital, as many people's food choices are heavily influenced by their emotional state. Emotional eating, for instance, is a common behavior where people consume food in response to stress, anxiety, or sadness, often leading to unhealthy eating patterns. On the other hand, specific nutrients and

foods have been scientifically proven to improve mood, focus, and relaxation. For instance, foods rich in omega-3 fatty acids, magnesium, and antioxidants can elevate mood and reduce stress. However, individuals are often unaware of the connection between their diet and their emotional well-being, resulting in poor food choices during times of stress or fatigue.

Given these factors, there is a clear necessity for a tool like **NUTRIFLEX**, which integrates personalized nutrition, gut health, and mood-based recommendations into one solution. Traditional meal planning apps may offer generic suggestions that don't account for individual biological needs, but NUTRIFLEX aims to change that. It considers both physical and emotional health in a holistic manner, providing users with personalized, scientifically-backed recommendations based on their unique health profiles.

The role of **Artificial Intelligence (AI)** in NUTRIFLEX is critical for optimizing meal recommendations. AI algorithms analyze user inputs such as Basal Metabolic Rate (BMR), Total Energy Expenditure (TEE), microbiome data, and mood to generate highly customized meal plans. The AI models can predict calorie content and offer smart recipe suggestions tailored to the user's health goals and emotional needs. By continuously learning from user behavior and feedback, AI enables NUTRIFLEX to evolve and provide increasingly accurate recommendations over time.

AI also helps in delivering **natural language processing (NLP)** capabilities, allowing users to input their health data and receive meal recommendations that are not just personalized but also easy to understand and follow. By leveraging large datasets and machine learning, NUTRIFLEX can offer users real-time insights into how their diet impacts both their physical health and emotional well-being, creating a truly dynamic health management experience.

3. SYSTEM DESIGN AND ARCHITECTURE

System Overview

The **NUTRIFLEX** application is built with a robust architecture designed to provide seamless interaction between the user and the AI-powered meal recommendation system. The architecture is modular, allowing different components to handle specific tasks such as user input, data processing, AI integration, and result generation. Below are the key components of the system:

Frontend: Streamlit Interface

The frontend of NUTRIFLEX is designed using **Streamlit**, an open-source Python library for creating web applications with minimal code. The goal of the frontend is to provide an intuitive and interactive user interface that makes it easy for users to input their health data, track their progress, and receive personalized meal recommendations.

- **User Input Forms:**
The frontend includes various forms where users can input their personal data such as weight, height, age, activity level, mood, and gut microbiome information. These inputs are crucial for generating personalized recommendations.
- **Navigation and Visualization:**
The Streamlit interface features a sidebar that allows users to navigate between different sections such as calorie prediction, gut health, and mood-based meal recommendations. It also includes charts and visualizations to help users track their health metrics over time, providing a clear and engaging experience.
- **Interactive Components:**
Users can select options, such as their current mood or dietary preferences, through dropdowns and radio buttons. This interactivity enhances the user experience and allows the app to dynamically update the AI-generated recommendations in real time.

The frontend is designed to be responsive and user-friendly, ensuring that users of all technical backgrounds can easily interact with the app.

Backend: AI Model Integration

The backend of NUTRIFLEX is where the core logic and AI models operate. It processes the user input, applies AI models for predictions and recommendations, and returns the results to the frontend. The backend architecture includes the following components:

- AI-Powered Prediction Models:**
NUTRIFLEX integrates **machine learning models** such as Random Forest, XGBoost, and Gradient Boosting to predict the calorie content of meals and to generate personalized meal recommendations. These models are trained on historical food and nutritional data to provide accurate calorie predictions based on the user's input.
- AI for Natural Language Processing (NLP):**
For more advanced, personalized responses, NUTRIFLEX uses GPT-3 model to generate meal recommendations based on user input such as microbiome data or mood. The model takes prompts generated from user data and formulates responses that cater to the user's health goals, mood, and gut health.
- Data Processing Logic:**
The backend handles complex data processing, including parsing user inputs, applying transformations, and feeding the data into the appropriate AI models. For example, it calculates Basal Metabolic Rate (BMR) and Total Energy Expenditure (TEE) based on the user's physical data before generating meal plans.
- Response Generation:**
Once the AI models have processed the input, the backend formulates responses such as calorie predictions or meal plans. These results are then sent back to the frontend, where they are displayed in a user-friendly format.

Databases: Data Storage and Retrieval

The system uses databases to store and manage user data, including past interactions and health metrics. These databases play a critical role in personalizing recommendations and ensuring that the system can learn and adapt over time.

- User Profiles:**
Each user has a unique profile in the database where personal details such as age, weight, height, and activity level are stored. This data is used for calculating BMR and TEE, which form the basis of calorie and meal recommendations.
- Microbiome Data:**
Users can input microbiome-related information, such as the presence of specific bacteria like Lactobacillus or Bifidobacterium. This data is stored in a dedicated section of the database and is used to create personalized gut health recommendations.
- Historical Data:**
The database stores historical data for each user, including past meal plans, calorie consumption, and mood states. This allows the AI models to track user behavior and improve the accuracy of future recommendations by learning from patterns in the data.

- **Gamification Data:**

Data related to the gamification system, such as points, badges, and rewards, is also stored. This keeps users engaged and motivated to follow through with their health goals.

The databases are secure and optimized for efficient retrieval, ensuring that users can quickly access their data and receive real-time updates to their health plans.

Data Flow

The flow of data through the system is critical to ensuring that NUTRIFLEX delivers timely and accurate recommendations. The following describes how data moves from the point of user input through the AI's recommendation engine.

1. **User Input Collection:**

- Users start by entering their health-related data through the frontend interface. This includes physical metrics (height, weight), dietary preferences, mood, and microbiome data. These inputs are collected through forms and interactive elements such as dropdowns and sliders.

2. **Data Validation and Preprocessing:**

- Before the data is processed, the backend validates the input to ensure that all fields are filled correctly. For example, it checks whether the user's weight falls within a healthy range or whether the mood selection is consistent with past inputs. After validation, the data is preprocessed (e.g., scaling, normalization) to ensure it can be correctly fed into machine learning models.

3. **AI Processing and Analysis:**

- The validated data is passed through different AI models depending on the type of recommendation. For example, if the user is looking for calorie predictions, the Random Forest model will be used to estimate the caloric content based on the food items and their associated nutritional data. If the user has entered microbiome information, the GPT-3 model will generate a personalized meal plan that considers the user's gut health.

4. **Meal Plan Generation:**

- Based on the AI's analysis, a detailed meal plan is created, which includes meal suggestions tailored to the user's health goals, gut health, and mood. For example, if the user reports feeling fatigued, the system might recommend a meal rich in omega-3 and magnesium to boost energy levels.

5. **Data Storage and Feedback Loop:**

- The generated recommendations, along with any new user input, are saved in the database for future reference. As the user continues to interact with the system, the AI models learn from these inputs, improving the accuracy of future recommendations.

6. **Display Results:**

- Finally, the personalized meal plan or calorie prediction is sent back to the frontend, where the user can view the results in an interactive and engaging format. Visual elements such as charts and graphs help users track their progress over time.

4. FEATURE BREAKDOWN

NUTRIFLEX offers a set of advanced, AI-driven features designed to provide personalized nutrition recommendations, calorie predictions, and mood-boosting suggestions. The following section provides a detailed technical explanation of the key features of the application, including the models and algorithms used for implementation.

Calorie Prediction

NUTRIFLEX’s **Calorie Prediction** feature employs machine learning models to accurately predict the caloric content of meals based on user-provided input such as ingredients, portion sizes, and meal types. Two key algorithms are used for this task: **Random Forest** and **XGBoost**.

- 1. **Predictive Models:**
 - **Random Forest:** This model is a robust ensemble learning method that builds multiple decision trees to make accurate predictions. It is particularly effective in handling high-dimensional datasets, which in this case include a wide array of food items and their corresponding nutritional values. Random Forest is used in NUTRIFLEX to predict the caloric content of meals by analyzing the nutritional properties of each food ingredient.
 - **XGBoost (Extreme Gradient Boosting):** XGBoost is another ensemble method that applies boosting techniques to improve the accuracy of calorie predictions. It works by combining multiple weak learners (decision trees) and iteratively refining their predictions to minimize error. XGBoost is highly efficient and allows NUTRIFLEX to provide quick, real-time calorie predictions.
 - **Gradient Boosting:** Gradient Boosting is a general boosting algorithm that builds models in a stage-wise fashion, optimizing the loss function with respect to the predictions of the previous models. Each subsequent model corrects the errors made by its predecessors, resulting in a strong overall model. In NUTRIFLEX, Gradient Boosting enhances the calorie prediction process by leveraging its ability to capture complex patterns in the data, ensuring more accurate caloric estimates based on user inputs.
- 2. **Input Features:**
 - **User-Specific Features:** Age, weight, height, gender, and activity level (to determine metabolic rate).
 - **Food-Specific Features:** Food type, portion size, macronutrient composition (carbohydrates, fats, proteins), and food preparation method.
 - **Nutritional Data:** A large dataset of foods with corresponding nutritional information, including calorie counts, is used as the training data for the machine learning models.
- 3. **Calorie Calculation:** Once the user enters their food details, the system processes the input through the trained models. Each food item is mapped to its corresponding nutritional values, and the model predicts the total calorie content based on the ingredients and portion sizes. The results are displayed to the user, allowing them to track their caloric intake.

Meal Planner

The **Meal Planner** feature provides users with meal recommendations that are tailored to their specific health needs. The system calculates two key metrics—**Basal Metabolic Rate (BMR)** and **Total Energy Expenditure (TEE)**—to create customized meal plans.

1. **BMR Calculation:** BMR is the number of calories a person needs to maintain basic physiological functions (e.g., breathing, circulation) while at rest. It is calculated using the **Mifflin-St Jeor Equation**:

$$\text{BMR} = 10 \times \text{weight(kg)} + 6.25 \times \text{height(cm)} - 5 \times \text{age} + \text{gender factor}$$
 where the gender factor is +5 for males and -161 for females.
2. **TEE Calculation:** TEE represents the total number of calories required per day, including physical activity. It is calculated by multiplying BMR with an **activity multiplier** (ranging from 1.2 for sedentary individuals to 1.9 for highly active individuals).
3. **Meal Recommendation Algorithm:** Based on the BMR and TEE calculations, NUTRIFLEX generates meal plans that align with the user's daily caloric needs. The app recommends meals that match the user's dietary preferences and nutritional goals (e.g., weight loss, muscle gain). The system ensures that the recommended meals provide a balanced intake of macronutrients (proteins, fats, and carbohydrates), tailored to the user's metabolic rate and energy expenditure.

Gut Health Analysis

NUTRIFLEX's **Gut Health Analysis** feature is designed to improve users' gut microbiota by providing personalized meal suggestions based on their microbiome data. This feature takes into account the user's gut flora and makes dietary recommendations aimed at enhancing gut health.

1. **Microbiome Data Input:** Users are asked to input information related to their gut microbiome, including the levels of beneficial bacteria such as **Lactobacillus** and **Bifidobacterium**. These bacteria play crucial roles in digestion, nutrient absorption, and immune function.
2. **AI-Powered Meal Recommendations:**
 - The app uses GPT-3 model for natural language processing (NLP) to create personalized meal suggestions based on microbiome data. GPT-3 processes the user's gut health information and generates meals that promote the growth of beneficial bacteria while minimizing the intake of foods that could disrupt gut balance (e.g., foods high in sugar or refined carbohydrates).
 - The generated meal plan includes foods rich in **prebiotics** (e.g., garlic, onions) and **probiotics** (e.g., yogurt, kefir), which support the growth of healthy bacteria. For users with a specific gut health issue, such as low levels of Lactobacillus, GPT-3 recommends meals that can help rebalance their gut flora.
3. **Gut Health Optimization:** By following these recommendations, users can optimize their gut health, which in turn improves digestion, boosts immunity, and enhances mental well-being, thanks to the gut-brain connection.

Mood Booster (Mind Enhancer)

The **Mood Booster** feature, also referred to as the **Mind Enhancer**, provides meal and activity recommendations based on the user's current mood. This feature is designed to improve the user's emotional well-being by offering scientifically-backed nutritional advice.

1. **Mood Input:** Users are prompted to enter their current mood (e.g., happy, sad, stressed, anxious, fatigued). This input serves as the starting point for generating mood-enhancing meal and activity suggestions.

2. Algorithm for Mood-Based Recommendations:

- The backend algorithm maps the user's mood state to specific nutrients and foods known to improve emotional health. For example:
 - **Stress:** Foods high in magnesium, such as dark leafy greens and nuts, can help reduce stress.
 - **Fatigue:** Foods rich in omega-3 fatty acids and complex carbohydrates can boost energy and focus.
 - **Anxiety:** Foods rich in antioxidants (e.g., berries, green tea) and omega-3s help reduce anxiety symptoms.
- 3. **Personalized Meal Suggestions:** Using the mood input, the AI models generate a list of foods or meals designed to enhance mood. For example, if a user indicates they are feeling stressed, the system might suggest a meal rich in magnesium (e.g., a spinach and avocado salad) to help relax the nervous system.
- 4. **Activity Suggestions:** In addition to meal recommendations, the Mind Enhancer feature also provides suggestions for activities (e.g., meditation, light exercise) that can further improve the user's mood and mental state.

5. AI INTEGRATION

This section provides a technical description of how AI models are integrated into the NUTRIFLEX application. We will cover the usage of the AI API for generating personalized meal plans, insights into predictive modeling for calorie predictions, and the implementation of natural language processing (NLP) for analyzing user inputs.

AI API Usage

The integration of the AI API is a pivotal aspect of NUTRIFLEX, enabling the application to generate personalized meal plans based on user inputs related to gut health and mood.

1. Prompt Creation:

- The application constructs a detailed prompt that encapsulates the user's microbiome data, current mood, and dietary preferences. This prompt serves as the input to the AI model, guiding it to produce relevant meal recommendations:

2. API Interaction:

- The app sends the constructed prompt to the AI API, specifying the model to be used (e.g., gpt-3.5-turbo). The interaction is achieved through an API call that processes the input and returns the AI-generated output:

```
model="gpt-3.5-turbo",
```

- The response from the API is parsed to extract the meal plan generated by the AI, which is then presented to the user.

3. Dynamic Meal Planning:

- The prompts are designed to be flexible, allowing the AI to account for various user scenarios. For example, if a user expresses a preference for vegetarian meals or has specific dietary restrictions, the prompt can be adjusted accordingly:

Predictive Modeling

NUTRIFLEX employs advanced machine learning models for calorie prediction, utilizing historical food data to provide accurate estimations based on user selections.

1. Model Selection:

- The application integrates several predictive models, including Random Forest, XGBoost, and Gradient Boosting. These models are well-suited for handling high-dimensional datasets, such as the nutritional properties of various food items.

2. Model Training:

- The training process involves collecting historical food data, which includes nutritional information and corresponding calorie counts. The data is preprocessed to ensure cleanliness and relevance, followed by splitting into training and testing datasets:

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(features, labels, test_size=0.2)
```

- Each model is trained using the training dataset, allowing it to learn the underlying patterns that correlate food characteristics with calorie content:

```
model = RandomForestRegressor()
```

```
model.fit(X_train, y_train)
```

3. Model Validation:

- After training, the models are validated using the test dataset to assess their predictive performance. Metrics such as Mean Absolute Error (MAE) and R-squared scores are calculated to evaluate accuracy:

```
from sklearn.metrics import mean_absolute_error, r2_score
```

```
predictions = model.predict(X_test)
```

```
print("MAE:", mean_absolute_error(y_test, predictions))
```

```
print("R-squared:", r2_score(y_test, predictions))
```

4. Real-Time Predictions:

- Once validated, the models are deployed to make real-time predictions based on user inputs. For instance, when a user specifies a food item and portion size, the model predicts the caloric content, which is displayed to the user:

```
calorie_prediction = model.predict([[food_item, portion_size]])
```

NLP and AI Interaction

Natural Language Processing (NLP) is utilized within NUTRIFLEX to enhance the interaction between users and the AI, ensuring that the application can effectively interpret and respond to user inputs.

1. Input Analysis:

- The app analyzes user inputs, such as mood selection or microbiome data, to determine the context and intent behind the input. This analysis is crucial for generating appropriate prompts for the OpenAI API:

```
user_input = st.text_input("Enter your dietary preferences or restrictions:")
```

2. Contextual Understanding:

- By employing NLP techniques, the application can identify keywords and sentiment in user inputs, allowing it to adjust its responses accordingly. For example, if a user indicates they are feeling stressed, the AI can prioritize recommendations that promote relaxation

3. Generating Responses:

- The output from the OpenAI API can also be refined using NLP techniques to ensure clarity and relevance. This might include rephrasing suggestions or tailoring recommendations to better fit user preferences.

6. CODE WALKTHROUGH

In this section, we will provide a detailed walkthrough of the Final.py file, focusing on key functions and sections integral to the NUTRIFLEX application. This overview will cover user authentication, sidebar navigation, input handling, meal plan generation, and calorie prediction.

Login and Authentication

The authentication process in NUTRIFLEX is managed through Streamlit's session state. This approach ensures that user data is secure and only accessible after proper authentication.

1. User Login:

- When users first access the app, they are prompted to enter their credentials (username and password). The login form is created using Streamlit's input fields:

```
username = st.text_input("Username")
```

```
password = st.text_input("Password", type="password")
```

- Upon submission, the app checks the provided credentials against a predefined user database (could be a CSV file, database, etc.):

```
if st.button("Login"):
```

```
    if username in user_db and password == user_db[username]["password"]:
```

```
        st.session_state["logged_in"] = True
```

```
    else:
```

```
        st.error("Invalid credentials")
```

- If the credentials are valid, the user is granted access to the main application features.

2. Session Management:

- Streamlit's session state is used to maintain the user's logged-in status throughout their session. This allows the app to render personalized content without requiring repeated logins:

if "logged_in" not in st.session_state:

```
st.session_state["logged_in"] = False
```

Sidebar Navigation

NUTRIFLEX incorporates a sidebar for intuitive navigation across various application modules. The sidebar is built using Streamlit's sidebar component, enabling users to switch between functionalities seamlessly.

1. Sidebar Structure:

- The sidebar is created with the `st.sidebar` module, which includes links to different features like meal planning, gut health analysis, mood enhancer, and user profile settings:

```
st.sidebar.title("NUTRIFLEX")
```

```
menu_options = ["Home", "Meal Planner", "Gut Health", "Mood Enhancer", "Profile"]
```

```
selected_option = st.sidebar.radio("Select an option:", menu_options)
```

2. Dynamic Content Rendering:

- Based on the user's selection, different functions are called to render the corresponding content:

```
if selected_option == "Meal Planner":
```

```
    meal_planner()
```

```
elif selected_option == "Gut Health":
```

```
    gut_health_analysis()
```

```
elif selected_option == "Mood Enhancer":
```

```
    mood_enhancer()
```

Input Handling

The application collects various user inputs necessary for generating personalized recommendations. Input handling is critical for ensuring that the app functions effectively based on user data.

1. Microbiome Data:

- Users are prompted to input their microbiome data, which may include specific bacteria levels (e.g., Lactobacillus, Bifidobacterium). Input fields are created for each parameter:

```
lactobacillus = st.number_input("Lactobacillus Level", min_value=0)
```

```
bifidobacterium = st.number_input("Bifidobacterium Level", min_value=0)
```

2. Mood Input:

- Users select their current mood from a dropdown or radio button selection:

```
mood = st.selectbox("How do you feel today?", ["Happy", "Sad", "Stressed", "Anxious", "Fatigued"])
```

3. Health Parameters:

- Additional health parameters such as age, weight, height, and activity level are collected using similar input fields. This data is critical for calculations related to calorie needs and meal recommendations:

```
age = st.number_input("Age", min_value=0)
```

```
weight = st.number_input("Weight (kg)", min_value=0)
```

```
height = st.number_input("Height (cm)", min_value=0)
```

```
activity_level = st.selectbox("Activity Level", ["Sedentary", "Light", "Moderate", "Active", "Very Active"])
```

4. Data Validation:

- The app includes checks to ensure that the inputs are valid and complete before proceeding to calculations or recommendations, reducing the likelihood of errors during processing.

Meal Plan Generation

The meal plan generation function is a core feature of NUTRIFLEX, leveraging user input and AI to provide personalized meal suggestions.

1. Generating AI Prompts:

- Based on the microbiome data and mood input, the app constructs a prompt for the AI API. This prompt summarizes the user's health status and dietary preferences.

2. API Interaction:

- The constructed prompt is sent to OpenAI's API, and the app waits for the AI-generated response.

3. Processing AI Responses:

- The response from the AI contains the recommended meal plan, which is parsed and displayed to the user:


```
meal_plan = response["choices"][0]["message"]["content"]

st.write("Here is your personalized meal plan:")

st.write(meal_plan)
```

Calorie Prediction

The calorie prediction feature utilizes historical food data and machine learning models to provide users with accurate caloric estimates based on their meal choices.

1. Model Integration:

- NUTRIFLEX integrates multiple predictive models (Random Forest, XGBoost, Gradient Boosting) to enhance prediction accuracy. Each model is trained on historical food data that includes nutritional values and calorie counts.

2. User Input Processing:

- The app gathers food-related user inputs, such as ingredients and portion sizes, and formats the data for prediction:

```
food_item = st.text_input("Enter food item")
```

```
portion_size = st.number_input("Portion size (grams)", min_value=1)
```

3. Making Predictions:

- When users submit their food selections, the app passes the input data to the trained models to predict the caloric content:

```
calorie_prediction = model.predict([[food_item, portion_size]])
```

```
st.write(f"Predicted Calories: {calorie_prediction[0]}")
```

4. Feedback and Display:

- The predicted calorie count is displayed to the user, allowing them to make informed dietary choices. The app also encourages users to log their meals for ongoing tracking and personalized recommendations.

Gamification System

To keep users engaged and motivated, NUTRIFLEX incorporates a **gamification system** that rewards users for following their meal plans and achieving health milestones.

1. Points System:

- Users earn points for completing daily tasks such as tracking their meals, updating their mood, and following the app's recommendations.
- Points are accumulated based on the consistency and accuracy with which users adhere to their health plans. For instance, users might earn 10 points for logging a meal and 25 points for reaching a weekly nutrition goal.

2. Badges and Rewards:

- As users accumulate points, they unlock rewards, which serve as visual indicators of progress. For example, a user who consistently follows their meal plan might unlock a "Health Guru" reward.
- The app also offers rewards such as personalized meal plans or discounts on health-related products as an incentive for staying on track.

3. Progress Tracking:

- The gamification system tracks users' progress over time, showing how their nutrition and mood have improved as they follow the app's recommendations.
- Users can set **goals** (e.g., losing weight, improving mood) and receive positive reinforcement through points, badges, and motivational messages.

7. USER INTERACTION AND UI/UX DESIGN

This section provides a technical overview of the user interface (UI) and user experience (UX) design of the NUTRIFLEX application, built using Streamlit. The focus is on how users interact with the application, beginning with the login process and progressing through various functionalities such as meal planning, gut health analysis, and mood enhancement.

User Journey from Logging in to Interacting with Features

1. Login Interface:

- Upon accessing the application, users are greeted with a clean, straightforward login page. The design features minimal distractions, allowing users to focus on entering their credentials.
- The interface includes two input fields for username and password, which are styled to enhance usability.

A prominent login button is available, along with feedback messages for unsuccessful login attempts, ensuring a smooth entry into the app.

2. Main Dashboard:

- Once logged in, users are redirected to the main dashboard, which serves as the central hub for navigation. The sidebar is displayed prominently, allowing users to select various functionalities with ease. The sidebar features radio buttons or a dropdown menu for intuitive navigation.

3. Input Forms:

- Each feature within the application presents users with tailored input forms that collect relevant data:
 - **Microbiome Data Input:** Users can enter specific bacteria levels (e.g., Lactobacillus, Bifidobacterium) using numerical input fields. These fields are designed with clear labels and placeholders to guide user input.
 - **Mood Selection:** The mood input is designed as a dropdown or radio button selection, simplifying user interaction by providing predefined options that users can easily select.

4. Feature Interactions:

- Upon selecting a feature from the sidebar, users are presented with interactive elements related to that functionality:
 - **Meal Planner:** After submitting microbiome data and mood, users receive personalized meal recommendations generated through AI. The interface displays the generated meal plan in a clear, readable format.
 - **Gut Health Analysis:** Users can view detailed analysis results based on their microbiome data. The app may include visual aids like graphs or charts to illustrate findings, enhancing comprehension.
 - **Mood Enhancer:** Users can interact with the mood enhancement feature, receiving tailored meal/activity suggestions based on their mood inputs. The results are displayed prominently, ensuring users can easily understand their options.

5. Navigation Panels:

- The sidebar navigation remains consistent across different features, allowing users to switch effortlessly between functionalities. Each section retains a familiar layout and design style, creating a cohesive experience.
- Users can also return to the home page or access their profile settings at any time, reinforcing the user-centric design of the application.

8. FUTURE WORK AND IMPROVEMENTS

The NUTRIFLEX application aims to provide a comprehensive solution for personalized nutrition management. As technology continues to evolve, several enhancements can be made to further improve user experience and expand the app's functionalities. Below are potential areas for future development:

Wearable Device Integration

1. **Activity Tracking:**

- Integrating wearable devices, such as smartwatches and fitness trackers, will allow NUTRIFLEX to collect real-time data on users' physical activities. This integration can enable the app to provide more personalized meal suggestions based on users' daily energy expenditure and activity levels.
- For example, if a user engages in a high-intensity workout, the app can suggest higher-calorie meals to replenish energy levels and support recovery.

2. **Health Monitoring:**

- Wearable devices can also provide valuable health metrics such as heart rate, sleep patterns, and stress levels. By analyzing this data, NUTRIFLEX can adjust meal recommendations and dietary plans to align with users' overall health status and wellness goals.

Voice Assistance

1. **Enhanced User Interaction:**

- Implementing a voice-based assistant within NUTRIFLEX can simplify user interactions, making it easier for individuals to input data and receive recommendations. Users could verbally share their dietary preferences, mood, or health conditions, and the assistant could respond with appropriate meal plans and suggestions.
- This feature will be especially beneficial for users who prefer hands-free operation or those with accessibility needs.

2. **Hands-Free Meal Logging:**

- Users could also utilize voice commands to log meals or physical activities, making tracking effortless and integrated into their daily routine.

Recipe Database Integration

1. **External API Connections:**

- Establishing connections with external APIs, such as Zomato and Swiggy, will allow NUTRIFLEX to offer smart recipe recommendations and facilitate meal ordering directly from the app.
- Users could receive personalized recipe suggestions based on their nutritional needs and preferences, and with a simple click, they could order ingredients or complete meals from local restaurants.

2. Diverse Recipe Options:

- By integrating a wider range of recipes, including dietary preferences such as vegan, gluten-free, and low-carb, NUTRIFLEX can cater to a broader audience and promote healthier eating habits.

Advanced AI Features

1. Improved Predictive Models:

- The implementation of more advanced AI models, such as deep learning algorithms, could refine predictions and recommendations based on user data patterns over time. These models could analyze a larger set of features to enhance the accuracy of calorie predictions and meal suggestions.
- Continuous learning from user feedback can also improve the AI's ability to adapt to individual preferences and evolving dietary needs.

2. Personalized Insights:

- Advanced AI capabilities can enable NUTRIFLEX to provide users with personalized insights and feedback on their eating habits, helping them to identify patterns, successes, and areas for improvement. This would foster a more engaging and proactive approach to nutrition management.

CODE :

```
import streamlit as st

from streamlit_option_menu import option_menu

import pandas as pd

import pickle

import matplotlib.pyplot as plt

import numpy as np

from sklearn.preprocessing import StandardScaler

import random

import google.generativeai as genai

import os

import sqlite3

import hashlib

import time

import plotly.graph_objects as go

from PIL import Image


# Configure API key for Google Generative AI

# Configure API key for Google Generative AI
```

```

os.environ["API_KEY"] = " API_KEY "

genai.configure(api_key=os.environ["API_KEY"])

# Generation configuration
generation_config = {
    "temperature": 1,
    "top_p": 0.95,
    "top_k": 64,
    "max_output_tokens": 8192,
    "response_mime_type": "text/plain",
}

model_path = os.path.join('E:', 'Nutrition assistance',
                           'models', 'random_forest_model.pkl')

# Function to load pre-saved models

def load_model(model_file):
    with open(model_file, 'rb') as file:
        model = pickle.load(file)
    return model

# Load your pre-saved models
rf_model = load_model(r'/app/models/random_forest_model.pkl')
xgb_model = load_model(r'/app/models/XGBoost.pkl')
gb_model = load_model(r'/app/models/gradient_boosting.pkl')
catboost_model = load_model(r'/app/models/cat_boost.pkl')

# Load the training data for preprocessing
# Update with your dataset path
data = pd.read_csv(r'/app/data/nndb_flat.csv')
data_cleaned = data.drop(
    columns=['ID', 'ShortDescrip', 'Descrip', 'CommonName', 'MfgName'])

```



```

df1 = pd.get_dummies(data_cleaned, columns=[
    'FoodGroup', 'ScientificName'], drop_first=True)
df1.fillna(0, inplace=True)

# Fit the scaler on the training data
scaler = StandardScaler()
X = df1.drop(columns=['Energy_kcal'])
X_scaled = scaler.fit_transform(X)

# Define each page as a function

# Database setup
conn = sqlite3.connect('/app/data/users.db', check_same_thread=False)
c = conn.cursor()

# Create users table if it doesn't exist
def create_users_table():
    c.execute("""CREATE TABLE IF NOT EXISTS users
        (user_id TEXT PRIMARY KEY, password TEXT, name TEXT, phone TEXT, email TEXT)""")

# Insert user into database
def add_user(user_id, password, name, phone, email):
    hashed_password = hashlib.sha256(password.encode()).hexdigest()
    c.execute('INSERT INTO users (user_id, password, name, phone, email) VALUES (?, ?, ?, ?, ?)',
        (user_id, hashed_password, name, phone, email))
    conn.commit()

# Verify if user exists and password is correct
def verify_user(user_id, password):
    hashed_password = hashlib.sha256(password.encode()).hexdigest()
    c.execute('SELECT * FROM users WHERE user_id = ? AND password = ?', (user_id, hashed_password))
    return c.fetchone()

```

Function to generate health tips using Google Generative AI

```
def generate_health_tips():
```

```
    prompt = "Give me some personalized health tip for maintaining a healthy lifestyle every day."
```

```
    # define model
```

```
    model = genai.GenerativeModel(model_name="gemini-1.5-flash",generation_config=generation_config,)
```

```
    chat_session = model.start_chat(history=[])
```

```
    response = chat_session.send_message(prompt)
```

```
    return response.text.strip()
```

Simulate recommendations based on user behavior

```
def get_personalized_recommendation(user_data):
```

```
    if user_data['calories'] > 2000:
```

```
        return "Consider reducing your calorie intake to meet your weight loss goals."
```

```
    elif user_data['steps'] < 8000:
```

```
        return "Try to increase your steps to meet your daily activity goal."
```

```
    else:
```

```
        return "You're on track! Keep up the good work."
```

Function to display dashboard after login

```
def show_dashboard(user_id):
```

```
    st.title(f"Welcome, {user_id}!")
```

```
    prompt4 = f"" generate Thought for the day based on nutrition""
```

```
    # define model
```

```
    model = genai.GenerativeModel(model_name="gemini-1.5-flash",generation_config=generation_config,)
```

```
    chat_session4 = model.start_chat(history=[])
```

```
    response4 = chat_session4.send_message(prompt4)
```

```
    return response4.text.strip()
```

```
def home():
```

```
    st.markdown(
```

```
"""
```

```
<style>
```

```
@import url('https://fonts.googleapis.com/css2?family=Roboto:wght@400;700&display=swap');
```

```
body {
```

```
    font-family: 'Roboto', sans-serif;
```

```
}
```

```
.stButton button {
```

```
    font-size: 50px; /* Adjusted font size for buttons */
```

```
    font-weight: bold;
```

```
}
```

```
h1, h2, h3, h4, h5, h6 {
```

```
    font-family: 'Roboto', sans-serif;
```

```
    font-size: 30px;
```

```
}
```

```
p, li {
```

```
    font-family: 'Roboto', sans-serif;
```

```
    font-size: 24px; /* Adjusted font size for markdown text */
```

```
}
```

```
</style>
```

```
"""
```

```
unsafe_allow_html=True
```

```
)
```

```
st.title("Welcome to NUTRIFLEX!")
```

```
# Introduction
```

```
st.write(
```

"In today's fast-paced world, maintaining a balanced diet is essential for overall health and well-being. "

"NUTRIFLEX is designed to make personalized nutrition accessible and engaging for everyone."

)

Image

st.image(r"/app/images/Wearble1.jpg", use_column_width=True) # Update with your image path

What We Offer Section

st.header("What We Offer")

st.subheader("Calorie Prediction")

st.write(

"Accurately estimate calorie content based on your nutritional input using advanced machine learning models."

)

st.subheader("Personalized Meal Planning")

st.write(

"Generate customized meal plans that align with your Basal Metabolic Rate (BMR) and Total Energy Expenditure (TEE), "

"ensuring you meet your dietary goals.")

st.subheader("Recipe Recommendations")

st.write(

"Discover delicious recipes that cater to your preferences, dietary restrictions, and caloric requirements."

)

st.subheader("Gamified Experience")

st.write(

"Engage with our interactive features that turn food tracking into a rewarding game, encouraging healthier eating habits."

)

st.subheader("AI-Based Recipe Suggestions for Specific Health Conditions")

```

st.write(
    "Receive AI-generated recipe suggestions designed for specific health conditions, such as managing stress and improving mood. "
    "Enjoy meals that promote relaxation, focus, and emotional balance."
)

st.subheader("Gut Health Meal Plans")
st.write(
    "Generate personalized meal plans based on your gut microbiome data. By understanding your unique gut health, "
    "we help you make dietary choices that improve digestion, support immunity, and optimize overall well-being."
)

# Our Mission Section
st.header("Our Mission")
st.write(
    "At NutriFLEX, our mission is to empower individuals to make informed dietary choices through the use of technology. "
    "We believe that everyone deserves access to personalized nutrition assistance that is both effective and enjoyable."
)

# Call to Action
st.write(
    "Use the navigation menu to explore different features of the app. "
    "Let's embark on this journey towards better nutrition together!"
)
st.markdown("---")
st.write("© 2024 NutriFLEX. All rights reserved.")

# BMR Calculation (Mifflin-St Jeor Equation)

def calculate_bmr(age, gender, weight, height):
    if gender == 'Male':
        bmr = 10 * weight + 6.25 * height - 5 * age + 5

```

```
else:
```

```
    bmr = 10 * weight + 6.25 * height - 5 * age - 161
```

```
return bmr
```

```
# Gamify Experience
```

```
def game_fy():
```

```
    # Extract food items (assuming there's a column named 'FoodName')
```

```
    food_items = data['ShortDescrip'].unique()
```

```
    # Create a title for the app
```

```
    st.title("Food Intake Game")
```

```
    # Checkbox selection for food items
```

```
    selected_foods = st.multiselect("Select your food intake:", food_items)
```

```
    # Reward system
```

```
    reward_points = 0
```

```
    if selected_foods:
```

```
        # Example: 10 points per food item selected
```

```
        reward_points = len(selected_foods) * 10
```

```
    # Display rewards
```

```
    st.write("You have selected:", selected_foods)
```

```
    st.success(f"Total Reward Points: {reward_points}")
```

```
    # Optional: Display a message based on the total points
```

```
    if reward_points > 0:
```

```
        st.success("Great job! Keep up the healthy eating!")
```

```
    else:
```

```
        st.warning("Please select some food items to earn rewards.")
```

Calculate TEE based on activity level

```
def calculate_tee(bmr, activity_level):
    activity_factors = {
        "sedentary": 1.2,
        "lightly active": 1.375,
        "moderately active": 1.55,
        "very active": 1.725,
        "super active": 1.9
    }
    return bmr * activity_factors.get(activity_level, 1.2)
```

Generate meal plan based on TEE

```
def generate_meal_plan(tee, dataset):
    daily_kcal = tee

    # Define a balanced macro distribution: 50% carbs, 20% protein, 30% fat
    total_protein = 0.2 * daily_kcal / 4
    total_fat = 0.3 * daily_kcal / 9
    total_carbs = 0.5 * daily_kcal / 4

    # Filter food items from dataset based on macronutrient content
    filtered_data = dataset[['ShortDescrip', 'Energy_kcal',
                             'Protein_g', 'Fat_g', 'Carb_g']].dropna()

    meal_plan = pd.DataFrame(
        columns=['ShortDescrip', 'Energy_kcal', 'Protein_g', 'Fat_g', 'Carb_g'])
    total_energy, total_protein_g, total_fat_g, total_carb_g = 0, 0, 0, 0

    # Iteratively add food items until the goal is reached
    for index, row in filtered_data.iterrows():
```

```

    if (total_energy < daily_kcal) and (total_protein_g < total_protein) and (total_fat_g < total_fat) and (total_carb_g
< total_carbs):

        meal_plan = pd.concat([meal_plan, row.to_frame().T], ignore_index=True)

        total_energy += row['Energy_kcal']

        total_protein_g += row['Protein_g']

        total_fat_g += row['Fat_g']

        total_carb_g += row['Carb_g']

    return meal_plan

# Function to calculate BMR and generate meal plan

def calculate_BMR():

    st.title("Meal Planner based on Physique")

    age = st.number_input("Enter your age:", min_value=10,
                           max_value=100, value=25)

    gender = st.selectbox("Select gender:", ['Male', 'Female'])

    weight = st.number_input("Enter weight (in kg):",
                              min_value=20.0, max_value=800.0)

    height = st.number_input("Enter height (in cm):",
                              min_value=100.0, max_value=700.0)

    activity_level = st.selectbox("Select activity level:",
                                   ['sedentary', 'lightly active', 'moderately active', 'very active', 'super active'])

    if st.button("Generate Meal Plan"):

        bmr = calculate_bmr(age, gender, weight, height)

        tee = calculate_tee(bmr, activity_level)

        # Check if TEE is valid before generating meal plan

        if tee <= 0:

            st.error("Calculated TEE value is invalid. Please check your input.")

            return

```



```

meal_plan = generate_meal_plan(tee, data)

st.write(f"Your BMR: {bmr:.2f} kcal")
st.write(f"Your TEE: {tee:.2f} kcal")
st.write("\n### Meal Plan based on your TEE:\n")

if meal_plan.empty:
    st.write("No meal plan could be generated based on the provided TEE.")
else:
    st.dataframe(meal_plan)

# Define a function to generate the recipe based on calorie input

def smart_recipe_recommendation():
    st.title("Smart Recipe Recommendation")

    # Input: Take desired calorie count from the user
    calorie_input = st.number_input(
        "Enter your desired calorie count:", min_value=100, max_value=2000, step=50, value=500)

    # Input: Take desired protein intake from the user
    protein_input = st.text_input("Enter your protein intake:")

    # Input: Take allergy (if their of certain ingredients)
    allergy_inputs = st.text_input("Enter allergy ingredient (if any): ")

    # Button to trigger recipe generation
    generate_button = st.button("Generate Recipe")

    # Function to generate the recipe based on calorie input
    def generate_recipe(calories, protein, allergy):

```

```
prompt = f"""
```

```
I want to generate a recipe for a healthy meal that contains around {calories} calories.
```

```
The meal should include the following:
```

- Protein: {protein}
- Vegetables: broccoli, spinach
- Healthy fats: olive oil or avocado
- Allergy:{allergy}

```
Provide a step-by-step recipe, including ingredients and instructions,excluding {allergy} ingredient and the estimated calorie breakdown.
```

```
"""
```

```
try:
```

```
# Define the generative model with the selected configuration
```

```
model = genai.GenerativeModel(
    model_name="gemini-1.5-flash",
    generation_config=generation_config,
)
```

```
# Start a chat session and send the prompt to the model
```

```
chat_session = model.start_chat(history=[])
```

```
response = chat_session.send_message(prompt)
```

```
# Return the response text if valid
```

```
return response.text if response else "No response from AI model. Please try again."
```

```
except Exception as e:
```

```
# Handle potential errors
```

```
return f"An error occurred: {str(e)}"
```

```
# Generate and display the recipe when the button is clicked
```

```
if generate_button:
```

```
if protein_input: # Ensure protein input is provided
```

```
recipe = generate_recipe(
```

```

        calorie_input, protein_input, allergy_inputs)

    st.write(recipe)

else:

    st.error("Please provide the protein intake value.")


def Smart_health():

    st.title("Generate AI-based recipe suggestions for specific health conditions")


    # Input : Take an update of patients past or present Health condition
    health_input = st.text_input("Enter the Health Condition of patient:")


    # Input : mood of end User
    stress_level_input = st.slider("Stress Level (1-10)", 1, 10)
    dietary_preferences_input = st.text_input("Dietary Preferences (e.g., vegan, vegetarian, gluten-free)", "")


    # Button to trigger recipe generation
    generate_button1 = st.button("Get Meal Suggestions and Recipe")


    def generate_health(health, stress_level, dietary_preferences):

        prompt1 = f"""You are a nutritionist specializing in dietary plans for people with {health}.
        Based on the user's preferences:
        Stress Level: {stress_level}
        Dietary Preferences: {dietary_preferences}

        Please create a healthy recipe that takes into account the necessary dietary restrictions for this condition.
        Make sure the recipe is nutritious, balanced, and easy to prepare. Include the ingredients and preparation steps.
        """


        # Define the generative model with the selected configuration
        model = genai.GenerativeModel(
            model_name="gemini-1.5-flash",
            generation_config=generation_config,

```

)

Start a chat session and send the prompt to the model

chat_session1 = model.start_chat(history=[])

response1 = chat_session1.send_message(prompt1)

return response1.text

if generate_button1:

recipe1 = generate_health(

health_input, stress_level_input, dietary_preferences_input)

st.write(recipe1)

def gut_bacterial_anal():

st.title("Generate a personalized meal plan based on gut microbiome data.")

st.write("The AI recommends foods to balance the gut flora, support digestion, and improve overall gut health.")

lactobacillus_input = st.selectbox(

"Lactobacillus Level", ["low", "normal", "high"])

bifidobacterium_input = st.selectbox(

"Bifidobacterium Level", ["low", "normal", "high"])

firmicutes_input = st.selectbox(

"Firmicutes Level", ["low", "normal", "high"])

bacteroidetes_input = st.selectbox(

"Bacteroidetes Level", ["low", "normal", "high"])

akkermansia_input = st.selectbox(

"Akkermansia Level", ["low", "normal", "high"])

clostridium_difficile_input = st.selectbox(

"Clostridium difficile Presence", ["present", "absent"])

generate_button2 = st.button("Generate Plan")

Function to call OpenAI API and generate meal plan

```
def generate(lactobacillus, bifidobacterium, firmicutes, bacteroidetes, akkermansia, clostridium_difficile):
```

```
    prompt2 = f"""
```

```
    You are a nutritionist specialized in gut health. Based on the following microbiome data:
```

```
    Lactobacillus: {lactobacillus}
```

```
    Bifidobacterium: {bifidobacterium}
```

```
    Firmicutes: {firmicutes}
```

```
    Bacteroidetes: {bacteroidetes}
```

```
    Akkermansia: {akkermansia}
```

```
    Clostridium difficile: {clostridium_difficile}
```

```
    Please create a personalized meal plan to improve gut health, focusing on:
```

1. Increasing beneficial bacteria like Lactobacillus and Bifidobacterium.
2. Reducing harmful bacteria like Clostridium difficile.
3. Enhancing microbial diversity.

```
    Include breakfast, lunch, dinner, and snacks, with detailed nutritional recommendations.
```

```
    """
```

```
    # Define the generative model with the selected configuration
```

```
    model = genai.GenerativeModel(
        model_name="gemini-1.5-flash",
        generation_config=generation_config,
    )
```

```
    # Start a chat session and send the prompt to the model
```

```
    chat_session2 = model.start_chat(history=[])
    response2 = chat_session2.send_message(prompt2)
```

```
    return response2.text
```

```
if generate_button2:
```

```
    # Call the generate_meal_plan function and display the result
```

```
    plan = generate(lactobacillus_input, bifidobacterium_input, firmicutes_input,
```

```
bacteroidetes_input, akkermansia_input, clostridium_difficile_input)
```

```
st.write(plan)
```

```
def calorie_prediction():
```

```
    st.title("Calorie Prediction")
```

```
    st.write(
```

```
        "This app predicts calorie content.")
```

```
    # User input section
```

```
    st.header("Input Nutritional Values")
```

```
    protein_g = st.number_input("Protein (g)", min_value=0.0, step=0.1)
```

```
    fat_g = st.number_input("Fat (g)", min_value=0.0, step=0.1)
```

```
    sugar_g = st.number_input("Sugar (g)", min_value=0.0, step=0.1)
```

```
    vitamin_a_mcg = st.number_input("Vitamin A (mcg)", min_value=0.0, step=1.0)
```

```
    vitamin_b6_mg = st.number_input(
```

```
        "Vitamin B6 (mg)", min_value=0.0, step=0.01)
```

```
    vitamin_b12_mcg = st.number_input(
```

```
        "Vitamin B12 (mcg)", min_value=0.0, step=0.1)
```

```
    vitamin_c_mg = st.number_input("Vitamin C (mg)", min_value=0.0, step=0.1)
```

```
    vitamin_e_mg = st.number_input("Vitamin E (mg)", min_value=0.0, step=0.01)
```

```
    thiamin_mg = st.number_input("Thiamin (mg)", min_value=0.0, step=0.01)
```

```
    calcium_mg = st.number_input("Calcium (mg)", min_value=0.0, step=1.0)
```

```
    copper_mcg = st.number_input("Copper (mcg)", min_value=0.0, step=0.1)
```

```
    iron_mg = st.number_input("Iron (mg)", min_value=0.0, step=0.01)
```

```
    magnesium_mg = st.number_input("Magnesium (mg)", min_value=0.0, step=0.1)
```

```
    manganese_mg = st.number_input("Manganese (mg)", min_value=0.0, step=0.01)
```

```
    # Prepare the input data for the model
```

```
    input_data = {
```

```
        'Protein_g': protein_g,
```

```
        'Fat_g': fat_g,
```

```
        'Sugar_g': sugar_g,
```

```

'VitaminA_mcg': vitamin_a_mcg,
'VitaminB6_mg': vitamin_b6_mg,
'VitaminB12_mcg': vitamin_b12_mcg,
'VitaminC_mg': vitamin_c_mg,
'VitaminE_mg': vitamin_e_mg,
'Thiamin_mg': thiamin_mg,
'Calcium_mg': calcium_mg,
'Copper_mcg': copper_mcg,
'Iron_mg': iron_mg,
'Magnesium_mg': magnesium_mg,
'Manganese_mg': manganese_mg
}

```

```

# Create a DataFrame from the input data

```

```

input_df = pd.DataFrame([input_data])

```

```

input_df = pd.get_dummies(input_df) # One-hot encode

```

```

# Match original features

```

```

input_df = input_df.reindex(columns=X.columns, fill_value=0)

```

```

# Prediction button

```

```

if st.button("Predict"):

```

```

    # Initialize prediction variables

```

```

    rf_prediction = xgb_prediction = gb_prediction = catboost_prediction = None

```

```

    # Predict using each model

```

```

    gb_prediction = gb_model.predict(input_df)[0]

```

```

    # Display predictions

```

```

    st.subheader("Predicted Calorie Content")

```

```

    st.write(f"Predicted Calorie: {gb_prediction:.2f} kcal")

```

```

def mental_health_insights():

```

```

st.title("Personalized Food and Activity Suggestions Based on Your Mood")

st.write("The AI recommends foods and activities that can help improve your mental well-being.")


# Collect mood score from the user

mood_input = st.selectbox(

    "How do you feel today? (Mood score)", ["Very Bad (1-2)", "Bad (3-4)", "Okay (5-6)", "Good (7-8)", "Very Good (9-10)"]

)

generate_button2 = st.button("Generate Suggestions")


# Function to call OpenAI API and generate suggestions
def generate_suggestions(mood_score):

    # Set the appropriate prompt based on mood_score

    if mood_score in ["Very Bad (1-2)", "Bad (3-4)"]:

        prompt = f"""

        The user is feeling down with a mood score of {mood_score}.

        Suggest comforting foods that can improve mental well-being, and activities that can help the user relax and feel better.

        """

    elif mood_score in ["Okay (5-6)"]:

        prompt = f"""

        The user has a mood score of {mood_score}.

        Suggest energizing and mood-boosting foods, and recommend activities that can uplift the user's spirits.

        """

    else:

        prompt = f"""

        The user is feeling great with a mood score of {mood_score}.

        Suggest foods and activities that can help maintain the user's positive mood and keep their energy high.

        """

# Define the generative model with the selected configuration

model = genai.GenerativeModel(

    model_name="gemini-1.5-flash",

```



```

generation_config=generation_config,)

# Start a chat session and send the prompt to the model
chat_session = model.start_chat(history=[])
response = chat_session.send_message(prompt)
return response.text

if generate_button2:
    suggestions = generate_suggestions(mood_input)
    st.write(suggestions)

def personalized_form():
    st.header("Display personalized meal suggestions based on user input.")
    # Filter relevant columns for analysis
    relevant_columns = ['ShortDescrip',
                        'Energy_kcal', 'Protein_g', 'Fat_g', 'Carb_g']
    filtered_data1 = data[relevant_columns]

    # Ensure numeric columns are converted properly
    filtered_data1['Energy_kcal'] = pd.to_numeric(
        filtered_data1['Energy_kcal'], errors='coerce')
    filtered_data1['Protein_g'] = pd.to_numeric(
        filtered_data1['Protein_g'], errors='coerce')
    filtered_data1['Fat_g'] = pd.to_numeric(
        filtered_data1['Fat_g'], errors='coerce')
    filtered_data1['Carb_g'] = pd.to_numeric(
        filtered_data1['Carb_g'], errors='coerce')

    # Total calorie intake calculation
    total_calories = filtered_data1['Energy_kcal'].sum()
    st.write(f"Total Calories Consumed: {total_calories}")

```

```

# User input for health goals

target_calories = st.number_input(
    "Enter your target daily calorie intake:", value=2000)

# Function to suggest meals based on current intake
def suggest_meals(current_calories, target_calories, filtered_data):
    if current_calories > target_calories:
        # Suggest lower-calorie meals
        suggestions = filtered_data[filtered_data['Energy_kcal'] < (
            target_calories / 3)]
    else:
        # Suggest higher-calorie meals for muscle gain
        suggestions = filtered_data[filtered_data['Energy_kcal'] > (
            target_calories / 3)]
    return suggestions

# Suggest meals based on user input
meal_suggestions = suggest_meals(
    total_calories, target_calories, filtered_data1)

# Display meal suggestions
st.subheader("Meal Suggestions Based on Your Intake")
st.write(meal_suggestions)

# Assuming nutrient_sums is already defined as you have in your code
nutrient_sums = filtered_data1[[
    'Energy_kcal', 'Protein_g', 'Fat_g', 'Carb_g']].sum()

# Create a Plotly bar chart
fig = go.Figure()

# Add data to the figure

```

```
fig.add_trace(go.Bar(
    x=nutrient_sums.index,
    y=nutrient_sums.values,
    marker_color='blue' # You can customize the color
))
```

```
# Update the layout of the figure
```

```
fig.update_layout(
    title="Nutrient Intake Overview",
    xaxis_title="Nutrients",
    yaxis_title="Total Intake",
    plot_bgcolor='rgba(255, 255, 255, 0.8)', # Transparent background
    paper_bgcolor='rgba(0, 0, 0, 0)', # No background color for paper
)
```

```
# Display the Plotly chart in Streamlit
```

```
st.plotly_chart(fig)
```

```
def about():
```

```
    st.markdown(
```

```
        """"
```

```
<style>
```

```
@import url('https://fonts.googleapis.com/css2?family=Roboto:wght@400;700&display=swap');
```

```
body {
```

```
    font-family: 'Roboto', sans-serif;
```

```
}
```

```
.stButton button {
```

```
    font-size: 50px; /* Adjusted font size for buttons */
```

```
    font-weight: bold;
```

```
}
```

```
h1, h2, h3, h4, h5, h6 {
    font-family: 'Roboto', sans-serif;
    font-size: 30px;
}
```

```
p, li {
    font-family: 'Roboto', sans-serif;
    font-size: 24px; /* Adjusted font size for markdown text */
}
```

```
</style>
""",
unsafe_allow_html=True
)
```

```
# Set the title of the app
st.title("About the Creator")
```

```
# Add the image (make sure to replace the image path accordingly)
image = Image.open(r"/app/images/A_professional-looking_portrait_of_a_tech-savvy_cr.webp")
st.image(image, caption="The Creator behind NUTRIFLEX", use_column_width=True)
```

```
# Write the About section
st.write("""
```

The creator behind **NutriFLEX** is a passionate innovator with expertise in data science, AI, and health technology. Dedicated to improving how people interact with nutrition and wellness, they combine machine learning, predictive analytics, and human-centered design to develop practical, user-friendly tools that make everyday life healthier and more manageable.

With a strong foundation in nutrition science and technology, the creator has worked on various AI-driven projects, including personalized

health tracking, AI-powered meal planners, and mental well-being integration. Their vision is to empower individuals with smart tools that

adapt to their unique needs, helping them achieve their health goals and maintain a balanced lifestyle.

****NutriFLEX**** reflects this mission

by offering an intuitive platform that adjusts to user goals, provides personalized meal suggestions, and integrates modern technologies like

voice assistance and wearables.

Beyond their work on health tech, the creator enjoys exploring the latest in food science, psychology, and fitness. They share their insights,

progress, and wellness tips with a vibrant community online.

Stay connected to learn more about cutting-edge health tech and AI-driven wellness solutions!

""")

st.header(" Follow the creator on social media:")

Add social media links

st.markdown("""

[![Instagram](https://img.shields.io/badge/Instagram-@py_rex_47-%23E4405F?style=for-the-badge&logo=instagram)](https://instagram.com/py_rex_47)

[![Twitter](https://img.shields.io/badge/Twitter-@Patil5432-%231DA1F2?style=for-the-badge&logo=twitter)](https://x.com/Patil5432)

[![LinkedIn](https://img.shields.io/badge/LinkedIn-YourLinkedInProfile-%230A66C2?style=for-the-badge&logo=linkedin)](https://www.linkedin.com/in/rahul-patil-4bb533209)

""")

Join Us Section

st.header("Join Us on Your NutriFLEX Journey")

st.write(

"Whether you're looking to lose weight, gain muscle, or simply eat healthier, NutriFlex is here to support you every step of the way. "

"Our user-friendly interface and expert-backed recommendations will help you achieve your nutrition goals."

)

Get in Touch Section

st.header("Get in Touch")

```

st.write(
    "We would love to hear from you! If you have any questions, feedback, or suggestions, please reach out to us at
    rp3252154@gmail.com.")

st.image("https://images.pexels.com/photos/7564224/pexels-photo-
7564224.jpeg?auto=compress&cs=tinysrgb&w=600",use_column_width=True) # Replace with your image URL

# Footer

st.markdown("---")

st.write("© 2024 NutriFLEX. All rights reserved.")


# Main App with Option Menu for Navigation

st.sidebar.image(r"/app/images/NUTRIFLEX.jpg",
               use_column_width=True) # Add your logo here


def main():

    # Initialize session state for login status
    if 'logged_in' not in st.session_state:
        st.session_state.logged_in = False

    create_users_table() # Ensure the users table is created

    if not st.session_state.logged_in:

        st.title("User Registration and Login")

        menu = ["Login", "Register"]

        choice = st.sidebar.selectbox("Menu", menu)

        if choice == "Register":

            st.subheader("Create a New Account")

            user_id = st.text_input("User ID", key="register_user_id")

            password = st.text_input("Password", type="password", key="register_password")

            password_confirm = st.text_input("Confirm Password", type="password", key="register_password_confirm")

            name = st.text_input("Name", key="register_name")

```

```
phone = st.text_input("Phone Number", key="register_phone")
```

```
email = st.text_input("Email", key="register_email")
```

```
if st.button("Register"):
```

```
    if password != password_confirm:
```

```
        st.error("Passwords do not match!")
```

```
    elif len(user_id) == 0 or len(password) == 0 or len(name) == 0 or len(phone) == 0 or len(email) == 0:
```

```
        st.error("All fields are required!")
```

```
    else:
```

```
        try:
```

```
            add_user(user_id, password, name, phone, email) # Pass all parameters
```

```
            st.success("Registration successful! You can now log in.")
```

```
        except sqlite3.IntegrityError:
```

```
            st.error("User ID already exists. Try a different one.")
```

```
elif choice == "Login":
```

```
    st.subheader("Login to Your Account")
```

```
    user_id = st.text_input("User ID", key="login_user_id")
```

```
    password = st.text_input("Password", type="password", key="login_password")
```

```
if st.button("Login"):
```

```
    if verify_user(user_id, password):
```

```
        st.session_state.logged_in = True
```

```
        st.session_state.user_id = user_id
```

```
        st.success(f"Welcome, {user_id}!")
```

```
        time.sleep(1) # Adding a small delay to simulate loading
```

```
    else:
```

```
        st.error("Invalid User ID or Password!")
```

```
if st.session_state.logged_in:
```

```
    show_dashboard(st.session_state.user_id)
```

```

# Page header on every page
st.sidebar.header(f"User: {st.session_state.user_id}")

with st.sidebar:
    selected = option_menu(
        menu_title="Main Menu",
        options=["Home", "Calorie Prediction", "Meal Planner", "Gut Health", "Mood Booster",
                "Smart Recipe Recommendation", "Smart Health", "Gamification", "Personalized Form", "About"], #
required
        icons=['house', 'calculator', 'book', 'clipboard-check', 'emoji-smile-fill',
                'bag', 'heart', 'controller', 'clipboard', 'info-circle'],
        menu_icon="app-indicator", default_index=0, orientation="vertical")

# Logout button
if st.button("Logout"):
    st.session_state.logged_in = False
    del st.session_state.user_id
    st.success("You have been logged out.")
    st.experimental_rerun() # Refresh the app to show the login screen

# Display the selected page
if selected == "Home":
    home()
elif selected == "Calorie Prediction":
    calorie_prediction()
elif selected == "Meal Planner":
    calculate_BMR()
elif selected == "Smart Recipe Recommendation":
    smart_recipe_recommendation()
elif selected == "Gut Health":
    gut_bacterial_anal()
elif selected == "Smart Health":
    Smart_health()
elif selected == "Mood Booster":

```



```

    mental_health_insights()
elif selected == "Gamification":
    game_fy()
elif selected == "Personalized Form":
    personalized_form()
elif selected == "About":
    about()
else:
    st.warning("Please log in to access features.")

# Run the main function
if __name__ == "__main__":
    main()

```

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